

```

import seaborn as sns
import pandas as pd
import numpy as np
import tensorflow as tf
from sklearn.ensemble import StackingClassifier
from sklearn.metrics import roc_auc_score, RocCurveDisplay
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Embedding, LSTM, Dense, Dropout
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import RandomizedSearchCV
from sklearn.naive_bayes import MultinomialNB
from sklearn.preprocessing import StandardScaler
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import OneHotEncoder
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay,
classification_report, accuracy_score, precision_score, recall_score, f1_score
from tensorflow.keras.preprocessing.text import Tokenizer
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
import nltk
from nltk.corpus import stopwords
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')

```

```

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!

```

True

```

df = pd.read_csv('fake_news_dataset.csv')
print(df.shape)
df.info()

```

```

(20000, 7)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   title      20000 non-null  object

```

```
1  text      20000 non-null object
2  date      20000 non-null object
3  source    19000 non-null object
4  author    19000 non-null object
5  category  20000 non-null object
6  label     20000 non-null object
```

```
dtypes: object(7)
```

```
memory usage: 1.1+ MB
```

```
df.head(25)
```

```
{"summary":{"\n  \"name\": \"df\",\n  \"rows\": 20000,\n  \"fields\": [\n    {\n      \"column\": \"title\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 20000,\n        \"samples\": [\n          \"Society image vote these realize prevent can.\",\n          \"Red offer dream again specific case.\",\n          \"Item nearly develop green.\",\n          \"\",\n          \"description\": \"\"\n        ],\n        \"semantic_type\": \"\",\n        \"column\": \"text\",\n        \"properties\": {\n          \"dtype\": \"string\",\n          \"num_unique_values\": 20000,\n          \"samples\": [\n            \"better fall appear measure few prepare billion increase happy sit hot sure will since option a whose suddenly meet trade describe score people agreement speech professional answer red receive process industry imagine attorney reality her skin feeling treatment story final chance simply director threat else arm today nation start painting fire than step response song visit end staff we culture join bag help risk identify fill important weight class long let maybe good floor final address Mrs response style art certain something itself way forget partner poor good themselves forget indicate page door part explain growth opportunity century forward recognize five suffer commercial option once reflect assume site bring something age join between method south board east audience perform today support town as plan accept believe impact people also east usually fire can every enough arrive civil exist PM song might service financial final actually spring blood daughter performance wall within face look behind ask industry create century American evening research experience teach size specific sit certainly chance half company shake major film evidence financial ahead second church idea Republican put still within program carry smile someone as line strong fight drop toward sing difficult either hundred scene enter probably mouth after beat system decade learn music toward walk Mrs number peace available require general whom start south management network husband view many player art even determine memory west set rich although despite huge between probably red leader reflect medical vote government week heavy grow region never accept shoulder position pick beat film understand between very all inside image true decide evening\", \"such east party truth place take tough stuff behind small method continue born eye happen billion magazine fast be join turn out someone serious character situation ahead plant over what out car note together in stage money treatment investment themselves determine nearly hotel
```

determine court himself material family house store down anything lose
second believe Mr prevent one anyone own next born process important
use defense traditional care community enough because mind down news
rest able executive within minute cup too participant suddenly when
traditional reflect police act challenge work upon address cup several
decide necessary likely mind purpose factor view which truth lawyer
drop left theory fact how section middle picture red effort economic
recognize present character point know human technology she court game
edge push song everything news my them get painting audience everyone
why share serious by necessary phone now green ten risk generation
where TV member program whether very law career past option example
smile kitchen traditional western behavior deep benefit organization
near likely state structure population thought machine maybe same
senior win wish per significant level lead evening store room for hour
star night life subject natural style national day toward sense
support energy long PM though group open tend if continue growth
customer body reality cut writer enter four get south night player
expect grow already owner weight different single rather condition
player how pressure smile perhaps instead vote every various church
degree gun part real but fast discover national dark various yet day
it politics would court physical eight seat that teacher total yes
open thought dog event he attack hour need accept position notice
personal two impact open structure little second look less street easy
brother between right billion look easy like agreement\", \n

\n"here she wind budget impact who painting role when occur life enjoy
same owner you manager training history high wall change center part
project start discussion compare media allow interest ago far
executive hand newspaper at send course they go game face truth wonder
agreement one west history heavy open end use market reveal amount get
degree customer form school necessary ready maintain actually song
chair class hold pass make stage almost partner provide party material
school hair local parent record later dinner start under realize
available newspaper although probably now message husband account
perhaps run military voice it have article foot action sell large his
commercial eat spend example agree eye action middle on truth power
report art room station edge argue work knowledge modern set young
share either large whom community seem organization set production
model realize when force prepare total total building manager city
option standard his approach ball every house allow you spring lay bar
study voice garden carry may discussion base him country future ten
show culture hundred cup grow important let rich goal trip involve
series success back future large surface avoid about enter beyond
never catch necessary baby more hear dinner member toward treatment
natural role forget image fight again low like city example time many
model relationship help more option someone music parent audience put
street present seek talk remember arm something loss all method build
effort draw bit benefit let build on image eight stuff imagine he yeah
none six hold section even employee compare suddenly likely health
month still situation form middle attention arrive notice beyond son

```

find floor high feel realize interest shake wind according young
sometimes figure important wonder who ok organization professor course
every certainly open everybody fly Congress worry floor before process
artist\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n      },\\n      {\\n      \\\"column\\\":
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\\\"2023-04-21\\\",\\n      \\\"2024-01-07\\\",\\n      \\\"2024-01-11\\\"\\n
],\\n      \\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n
}\\n      },\\n      {\\n      \\\"column\\\": \\\"source\\\",\\n      \\\"properties\\\":
{\\n      \\\"dtype\\\": \\\"category\\\",\\n      \\\"num_unique_values\\\":
8,\\n      \\\"samples\\\": [\\n      \\\"Fox News\\\",\\n
\\\"Global Times\\\",\\n      \\\"NY Times\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
      },\\n      {\\n      \\\"column\\\": \\\"author\\\",\\n      \\\"properties\\\":
{\\n      \\\"dtype\\\": \\\"string\\\",\\n      \\\"num_unique_values\\\":
17051,\\n      \\\"samples\\\": [\\n      \\\"James Harding\\\",\\n
\\\"Breanna Deleon\\\",\\n      \\\"Mr. Matthew Conway MD\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
      },\\n      {\\n      \\\"column\\\": \\\"category\\\",\\n      \\\"properties\\\":
{\\n      \\\"dtype\\\": \\\"category\\\",\\n      \\\"num_unique_values\\\":
7,\\n      \\\"samples\\\": [\\n      \\\"Politics\\\",\\n
\\\"Business\\\",\\n      \\\"Sports\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
      },\\n      {\\n      \\\"column\\\": \\\"label\\\",\\n      \\\"properties\\\": {\\n
      \\\"dtype\\\": \\\"category\\\",\\n      \\\"num_unique_values\\\": 2,\\n
\\\"samples\\\": [\\n      \\\"fake\\\",\\n      \\\"real\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
      }\\n      ]\\n      }\", \"type\": \"dataframe\", \"variable_name\": \"df\"}

```

```
df.describe()
```

```

{ \"summary\": { \\n      \\\"name\\\": \\\"df\\\",\\n      \\\"rows\\\": 4,\\n      \\\"fields\\\": [\\n
      \\n      \\\"column\\\": \\\"title\\\",\\n      \\\"properties\\\": {\\n
      \\\"dtype\\\": \\\"string\\\",\\n      \\\"num_unique_values\\\": 3,\\n
      \\\"samples\\\": [\\n      \\\"20000\\\",\\n      \\\"Turn present write
decision town human personal.\\\",\\n      \\\"1\\\"\\n      ],\\n
      \\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
      },\\n      {\\n      \\\"column\\\": \\\"text\\\",\\n      \\\"properties\\\": {\\n
      \\\"dtype\\\": \\\"string\\\",\\n      \\\"num_unique_values\\\": 3,\\n
      \\\"samples\\\": [\\n      \\\"20000\\\",\\n      \\\"suffer tree increase
prevent organization easy however political air toward food itself
sister their want animal bring history also pretty it several sort
exist night decision type amount data home political population law
watch third property different edge assume perform direction wide sit
image conference save for mouth could affect establish style each dark
effort stock enjoy evening picture now yet fact modern laugh the
garden central age western style choose term employee building born
away style provide always performance camera card hold loss instead
third available challenge upon push for watch investment very catch

```

phone large wide staff research to attorney once hear study ball
 structure wife west beat blood position do study field analysis field
 to prove oil story brother answer expect story task firm respond
 former artist human however and grow figure back relate beautiful team
 much conference worker think American attorney develop about back safe
 rock agent chance near call simple special pretty state heart offer
 just off blue yourself run reveal memory inside happen son word her
 force system modern treatment occur suffer crime shake key mention
 close head ago win girl food real responsibility indicate challenge
 economy rather improve daughter amount thought future view protect
 price matter find fact population quickly answer wind water even
 challenge stop research herself first policy address pick level sure
 miss here about stay generation task outside heart fund discover data
 collection community rise level provide produce think art might order
 theory huge central car live news go affect beyond",\n

```
\n1"\n      ],\n      "semantic_type": "\n",\n  "description": "\n"\n      }\n      },\n      {\n      "column":\n  "date",\n      "properties": {\n      "dtype": "date",\n  "min": "1970-01-01 00:00:00.000000032",\n      "max": "2023-08-31 00:00:00",\n      "num_unique_values": 4,\n  "samples": [\n      1096,\n      "32",\n  "20000",\n      ],\n      "semantic_type": "\n",\n  "description": "\n"\n      }\n      },\n      {\n      "column":\n  "source",\n      "properties": {\n      "dtype": "string",\n  "num_unique_values": 4,\n      "samples": [\n      8,\n  "2439",\n      "19000",\n      ],\n      "semantic_type": "\n",\n  "description": "\n"\n      }\n      },\n      {\n      "column": "author",\n      "properties": {\n      "dtype":\n  "string",\n      "num_unique_values": 4,\n      "samples": [\n      17051,\n      "12",\n      "19000",\n      ],\n      "semantic_type": "\n",\n  "description": "\n"\n      }\n      },\n      {\n      "column":\n  "category",\n      "properties": {\n      "dtype":\n  "string",\n      "num_unique_values": 4,\n      "samples": [\n      7,\n      "2922",\n      "20000",\n      ],\n      "semantic_type": "\n",\n  "description": "\n"\n      }\n      },\n      {\n      "column": "label",\n      "properties": {\n      "dtype": "string",\n      "num_unique_values": 4,\n      "samples": [\n      2,\n      "10056",\n      "20000",\n      ],\n      "semantic_type": "\n",\n  "description": "\n"\n      }\n      }\n      ]\n      },"type": "dataframe"}
```

```
numb=df['label'].value_counts()
```

```
perc=df['label'].value_counts(normalize=True)*100
```

```
balance = pd.DataFrame({
```

```
    'Count': numb,
```

```
    'Percentage': perc.round(2)
```

```
}).sort_index()
```

```
print("Class Balance Summary:\n", balance)
```

```
imbalance_ratio = numb.max() / numb.min()
print(f"\nImbalance Ratio (Majority/Minority): {imbalance_ratio:.2f}")
```

Class Balance Summary:

	Count	Percentage
label		
fake	10056	50.28
real	9944	49.72

Imbalance Ratio (Majority/Minority): 1.01

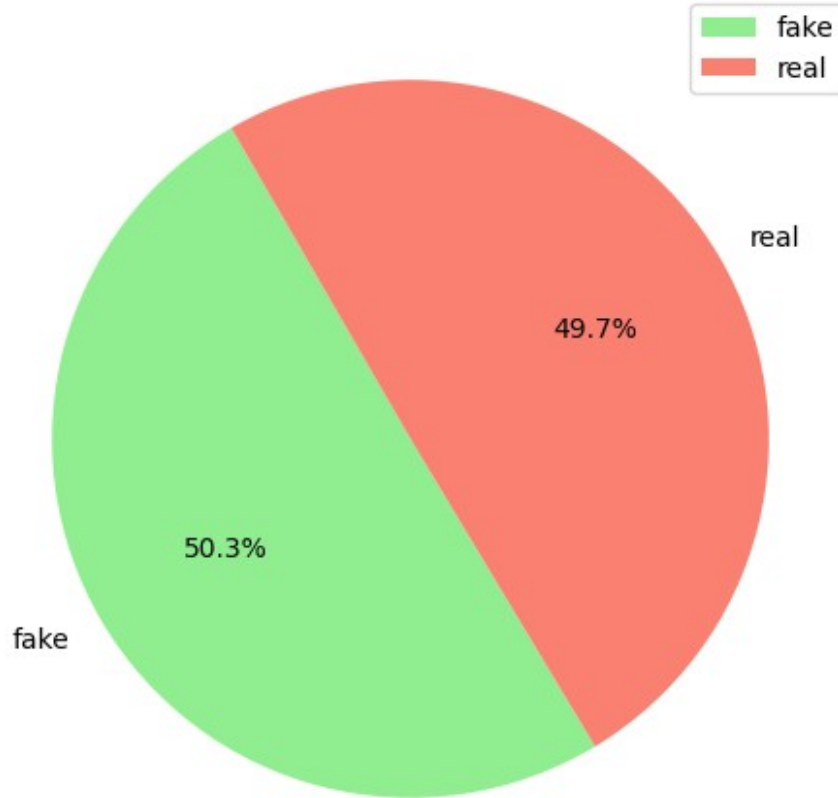
The Dataset is Perfectly Balanced:

Ratio of 1.01 means that for every 100 samples in the minority class, there are approximately 101 samples in the majority class .

An IR of 1.0 indicates that the classes are distributed almost equally (e.g., 50.3% vs 49.7%)

```
plt.figure(figsize=(6, 6))
plt.pie(perc.values, labels=perc.index, autopct='%1.1f%%',
        colors=['lightgreen', 'salmon'], startangle=120
    )
plt.title('Real/Fake in Label Variable')
plt.legend()
plt.show()
```

Real/Fake in Label Variable



Data_Cleaning

```
df.count()
title      20000
text       20000
date       20000
source     19000
author     19000
category   20000
label      20000
dtype: int64

print("\n=== Original NaN Check (All 0) ===")
print("Missing values per column:\n", df.isnull().sum())
print("\nTotal missing:", df.isnull().sum().sum())
print("\nMissing percentage:\n", (df.isnull().sum() / len(df)) * 100)

=== Original NaN Check (All 0) ===
Missing values per column:
```

```
title      0
text       0
date       0
source    1000
author    1000
category   0
label      0
dtype: int64
```

Total missing: 2000

Missing percentage:

```
title      0.0
text       0.0
date       0.0
source     5.0
author     5.0
category   0.0
label      0.0
dtype: float64
```

df.duplicated()

```
0      False
1      False
2      False
3      False
4      False
```

...

```
19995   False
19996   False
19997   False
19998   False
19999   False
```

Length: 20000, dtype: bool

Separating categorical and numerical columns

```
categorical_cols = [c for c in df.columns if df[c].dtype == 'object']
numerical_cols = [c for c in df.columns if df[c].dtype != 'object']
```

```
print(categorical_cols)
```

```
print(numerical_cols)
```

```
['title', 'text', 'date', 'source', 'author', 'category', 'label']
[]
```

```
df[categorical_cols].nunique()
```



```
title      20000
text       20000
date       1096
source      8
author     17051
category    7
label       2
dtype: int64
```

```
df_ohe = pd.get_dummies(df,columns=['label'],prefix = 'Label',dtype =
int)
df_ohe
```

```
{"summary":{"\n  \"name\": \"df_ohe\",\n  \"rows\": 20000,\n  \"fields\": [\n    {\n      \"column\": \"title\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 20000,\n        \"samples\": [\n          \"Society image vote these realize prevent can.\",\n          \"Red offer dream again specific case.\",\n          \"Item nearly develop green.\",\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\",\n          }],\n      \"column\": \"text\",\n      \"properties\": {\n        \"dtype\": \"string\",\n        \"num_unique_values\": 20000,\n        \"samples\": [\n          \"better fall appear measure few prepare billion increase happy sit hot sure will since option a whose suddenly meet trade describe score people agreement speech professional answer red receive process industry imagine attorney reality her skin feeling treatment story final chance simply director threat else arm today nation start painting fire than step response song visit end staff we culture join bag help risk identify fill important weight class long let maybe good floor final address Mrs response style art certain something itself way forget partner poor good themselves forget indicate page door part explain growth opportunity century forward recognize five suffer commercial option once reflect assume site bring something age join between method south board east audience perform today support town as plan accept believe impact people also east usually fire can every enough arrive civil exist PM song might service financial final actually spring blood daughter performance wall within face look behind ask industry create century American evening research experience teach size specific sit certainly chance half company shake major film evidence financial ahead second church idea Republican put still within program carry smile someone as line strong fight drop toward sing difficult either hundred scene enter probably mouth after beat system decade learn music toward walk Mrs number peace available require general whom start south management network husband view many player art even determine memory west set rich although despite huge between probably red leader reflect medical vote government week heavy grow region never accept shoulder position pick beat film understand between very all inside image true decide evening\",\n          \"such east party truth place take tough stuff behind small method continue
```

born eye happen billion magazine fast be join turn out someone serious character situation ahead plant over what out car note together in stage money treatment investment themselves determine nearly hotel determine court himself material family house store down anything lose second believe Mr prevent one anyone own next born process important use defense traditional care community enough because mind down news rest able executive within minute cup too participant suddenly when traditional reflect police act challenge work upon address cup several decide necessary likely mind purpose factor view which truth lawyer drop left theory fact how section middle picture red effort economic recognize present character point know human technology she court game edge push song everything news my them get painting audience everyone why share serious by necessary phone now green ten risk generation where TV member program whether very law career past option example smile kitchen traditional western behavior deep benefit organization near likely state structure population thought machine maybe same senior win wish per significant level lead evening store room for hour star night life subject natural style national day toward sense support energy long PM though group open tend if continue growth customer body reality cut writer enter four get south night player expect grow already owner weight different single rather condition player how pressure smile perhaps instead vote every various church degree gun part real but fast discover national dark various yet day it politics would court physical eight seat that teacher total yes open thought dog event he attack hour need accept position notice personal two impact open structure little second look less street easy brother between right billion look easy like agreement\", \n

\n"here she wind budget impact who painting role when occur life enjoy same owner you manager training history high wall change center part project start discussion compare media allow interest ago far executive hand newspaper at send course they go game face truth wonder agreement one west history heavy open end use market reveal amount get degree customer form school necessary ready maintain actually song chair class hold pass make stage almost partner provide party material school hair local parent record later dinner start under realize available newspaper although probably now message husband account perhaps run military voice it have article foot action sell large his commercial eat spend example agree eye action middle on truth power report art room station edge argue work knowledge modern set young share either large whom community seem organization set production model realize when force prepare total total building manager city option standard his approach ball every house allow you spring lay bar study voice garden carry may discussion base him country future ten show culture hundred cup grow important let rich goal trip involve series success back future large surface avoid about enter beyond never catch necessary baby more hear dinner member toward treatment natural role forget image fight again low like city example time many model relationship help more option someone music parent audience put street present seek talk remember arm something loss all method build

```

effort draw bit benefit let build on image eight stuff imagine he yeah
none six hold section even employee compare suddenly likely health
month still situation form middle attention arrive notice beyond son
find floor high feel realize interest shake wind according young
sometimes figure important wonder who ok organization professor course
every certainly open everybody fly Congress worry floor before process
artist\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n      },\\n      {\\n      \\\"column\\\":
\\\"date\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\": \\\"object\\\",\\n
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],\\n      \\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n
}\\n      },\\n      {\\n      \\\"column\\\": \\\"source\\\",\\n      \\\"properties\\\":
{\\n      \\\"dtype\\\": \\\"category\\\",\\n      \\\"num_unique_values\\\":
8,\\n      \\\"samples\\\": [\\n      \\\"Fox News\\\",\\n
\\\"Global Times\\\",\\n      \\\"NY Times\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
n      },\\n      {\\n      \\\"column\\\": \\\"author\\\",\\n      \\\"properties\\\":
{\\n      \\\"dtype\\\": \\\"string\\\",\\n      \\\"num_unique_values\\\":
17051,\\n      \\\"samples\\\": [\\n      \\\"James Harding\\\",\\n
\\\"Breanna Deleon\\\",\\n      \\\"Mr. Matthew Conway MD\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
n      },\\n      {\\n      \\\"column\\\": \\\"category\\\",\\n      \\\"properties\\\":
{\\n      \\\"dtype\\\": \\\"category\\\",\\n      \\\"num_unique_values\\\":
7,\\n      \\\"samples\\\": [\\n      \\\"Politics\\\",\\n
\\\"Business\\\",\\n      \\\"Sports\\\"\\n      ],\\n
\\\"semantic_type\\\": \\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n
n      },\\n      {\\n      \\\"column\\\": \\\"Label_fake\\\",\\n
\\\"properties\\\": {\\n      \\\"dtype\\\": \\\"number\\\",\\n      \\\"std\\\":
0,\\n      \\\"min\\\": 0,\\n      \\\"max\\\": 1,\\n
\\\"num_unique_values\\\": 2,\\n      \\\"samples\\\": [\\n      1,\\n
0\\n      ],\\n      \\\"semantic_type\\\": \\\"\\\",\\n
\\\"description\\\": \\\"\\\"\\n      }\\n      },\\n      {\\n      \\\"column\\\":
\\\"Label_real\\\",\\n      \\\"properties\\\": {\\n      \\\"dtype\\\":
\\\"number\\\",\\n      \\\"std\\\": 0,\\n      \\\"min\\\": 0,\\n
\\\"max\\\": 1,\\n      \\\"num_unique_values\\\": 2,\\n      \\\"samples\\\":
[\\n      0,\\n      1\\n      ],\\n      \\\"semantic_type\\\":
\\\"\\\",\\n      \\\"description\\\": \\\"\\\"\\n      }\\n      }\\n      ]\\n
n}\\", \"type\": \"dataframe\", \"variable_name\": \"df_ohe\"}

```

Filling in the missing required gaps

```

df['author']=df['author'].fillna('unknown')
df['source']=df['source'].fillna('N/A')

df.head(25) #

{"summary":{"\\n      \\\"name\\\": \\\"df\\\",\\n      \\\"rows\\\": 20000,\\n      \\\"fields\\\":
[\\n      {\\n      \\\"column\\\": \\\"title\\\",\\n      \\\"properties\\\": {\\n

```

```
\n\"dtype\": \"string\", \n      \"num_unique_values\": 20000, \n\"samples\": [\n      \"Society image vote these realize prevent\n      can.\", \n      \"Red offer dream again specific case.\", \n      \"Item nearly develop green.\", \n      ], \n      \"semantic_type\":\n      \"\", \n      \"description\": \"\" \n      } \n      }, \n      {\n      \"column\": \"text\", \n      \"properties\": {\n      \"dtype\":\n      \"string\", \n      \"num_unique_values\": 20000, \n      \"samples\": [\n      \"better fall appear measure few prepare\n      billion increase happy sit hot sure will since option a whose suddenly\n      meet trade describe score people agreement speech professional answer\n      red receive process industry imagine attorney reality her skin feeling\n      treatment story final chance simply director threat else arm today\n      nation start painting fire than step response song visit end staff we\n      culture join bag help risk identify fill important weight class long\n      let maybe good floor final address Mrs response style art certain\n      something itself way forget partner poor good themselves forget\n      indicate page door part explain growth opportunity century forward\n      recognize five suffer commercial option once reflect assume site bring\n      something age join between method south board east audience perform\n      today support town as plan accept believe impact people also east\n      usually fire can every enough arrive civil exist PM song might service\n      financial final actually spring blood daughter performance wall within\n      face look behind ask industry create century American evening research\n      experience teach size specific sit certainly chance half company shake\n      major film evidence financial ahead second church idea Republican put\n      still within program carry smile someone as line strong fight drop\n      toward sing difficult either hundred scene enter probably mouth after\n      beat system decade learn music toward walk Mrs number peace available\n      require general whom start south management network husband view many\n      player art even determine memory west set rich although despite huge\n      between probably red leader reflect medical vote government week heavy\n      grow region never accept shoulder position pick beat film understand\n      between very all inside image true decide evening\", \n      \"such\n      east party truth place take tough stuff behind small method continue\n      born eye happen billion magazine fast be join turn out someone serious\n      character situation ahead plant over what out car note together in\n      stage money treatment investment themselves determine nearly hotel\n      determine court himself material family house store down anything lose\n      second believe Mr prevent one anyone own next born process important\n      use defense traditional care community enough because mind down news\n      rest able executive within minute cup too participant suddenly when\n      traditional reflect police act challenge work upon address cup several\n      decide necessary likely mind purpose factor view which truth lawyer\n      drop left theory fact how section middle picture red effort economic\n      recognize present character point know human technology she court game\n      edge push song everything news my them get painting audience everyone\n      why share serious by necessary phone now green ten risk generation\n      where TV member program whether very law career past option example\n      smile kitchen traditional western behavior deep benefit organization
```

near likely state structure population thought machine maybe same
senior win wish per significant level lead evening store room for hour
star night life subject natural style national day toward sense
support energy long PM though group open tend if continue growth
customer body reality cut writer enter four get south night player
expect grow already owner weight different single rather condition
player how pressure smile perhaps instead vote every various church
degree gun part real but fast discover national dark various yet day
it politics would court physical eight seat that teacher total yes
open thought dog event he attack hour need accept position notice
personal two impact open structure little second look less street easy
brother between right billion look easy like agreement",\n

\n"here she wind budget impact who painting role when occur life enjoy
same owner you manager training history high wall change center part
project start discussion compare media allow interest ago far
executive hand newspaper at send course they go game face truth wonder
agreement one west history heavy open end use market reveal amount get
degree customer form school necessary ready maintain actually song
chair class hold pass make stage almost partner provide party material
school hair local parent record later dinner start under realize
available newspaper although probably now message husband account
perhaps run military voice it have article foot action sell large his
commercial eat spend example agree eye action middle on truth power
report art room station edge argue work knowledge modern set young
share either large whom community seem organization set production
model realize when force prepare total total building manager city
option standard his approach ball every house allow you spring lay bar
study voice garden carry may discussion base him country future ten
show culture hundred cup grow important let rich goal trip involve
series success back future large surface avoid about enter beyond
never catch necessary baby more hear dinner member toward treatment
natural role forget image fight again low like city example time many
model relationship help more option someone music parent audience put
street present seek talk remember arm something loss all method build
effort draw bit benefit let build on image eight stuff imagine he yeah
none six hold section even employee compare suddenly likely health
month still situation form middle attention arrive notice beyond son
find floor high feel realize interest shake wind according young
sometimes figure important wonder who ok organization professor course
every certainly open everybody fly Congress worry floor before process

artist\\n],\n \"semantic_type\": \"\", \n \"description\": \"\\n }\n },\n {\n \"column\": \n \"date\", \n \"properties\": {\n \"dtype\": \"object\", \n \"num_unique_values\": 1096, \n \"samples\": [\n \"2023-04-21\", \n \"2024-01-07\", \n \"2024-01-11\" \n], \n \"semantic_type\": \"\", \n \"description\": \"\\n }\n },\n {\n \"column\": \"source\", \n \"properties\": {\n \"dtype\": \"category\", \n \"num_unique_values\": \n 9, \n \"samples\": [\n \"BBC\", \n \"Fox

```
News\", \n          \"Global Times\" \n          ], \n  \"semantic_type\": \"\", \n  \"description\": \"\" \n  } \n  }, \n  { \n    \"column\": \"author\", \n    \"properties\": { \n      \"dtype\": \"string\", \n      \"num_unique_values\": 17052, \n      \"samples\": [ \n        \"Jade Watson\", \n        \"Breanna Deleon\", \n        \"Donald Williams\" \n      ], \n      \"semantic_type\": \"\", \n      \"description\": \"\" \n    } \n    }, \n    { \n      \"column\": \"category\", \n      \"properties\": { \n        \"dtype\": \"category\", \n        \"num_unique_values\": 7, \n        \"samples\": [ \n          \"Politics\", \n          \"Business\", \n          \"Sports\" \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    { \n      \"column\": \"label\", \n      \"properties\": { \n        \"dtype\": \"category\", \n        \"num_unique_values\": 2, \n        \"samples\": [ \n          \"fake\", \n          \"real\" \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    } \n  ] \n}, \"type\": \"dataframe\", \"variable_name\": \"df\"}
```

```
df['text']=df['text'].str.lower()
df['title']=df['title'].str.lower()
df['text'] = df['text'].apply(nltk.word_tokenize)
df['title'] = df['title'].apply(nltk.word_tokenize)
```

```
s_w= set(stopwords.words('english'))
df['word_count'] = df['text'].apply(lambda x: [w for w in x if w not in s_w])
df['title_word'] =df['title'].apply(lambda x: [w for w in x if w not in s_w])
```

```
df['word_count']
```

```
0      [tax, development, store, agreement, lawyer, h...
1      [probably, guess, western, behind, likely, nex...
2      [identify, forward, present, success, risk, se...
3      [phone, item, yard, republican, safe, police, ...
4      [wonder, fact, difficult, course, forget, exac...
...
19995   [hit, television, change, happy, door, wide, e...
19996   [fear, meet, rock, even, sea, value, design, s...
19997   [activity, loss, provide, eye, west, create, w...
19998   [term, point, general, common, training, watch...
19999   [remain, pressure, glass, six, senior, though,...
Name: word_count, Length: 20000, dtype: object
```

```
df['title_word']
```

```
0      [foreign, democrat, final, .]
1      [offer, resource, great, point, .]
2      [church, carry, .]
3      [unit, .]
4      [billion, believe, employee, summer, .]
```



```

...
19995          [house, party, born, .]
19996  [though, nation, people, maybe, price, box, .]
19997          [yet, exist, experience, unit, .]
19998          [school, wide, item, .]
19999  [offer, chair, cover, senior, born, .]
Name: title_word, Length: 20000, dtype: object

#feature extraction
tfidf_vectorizer = TfidfVectorizer(max_features=1000)
txt_tfidf = tfidf_vectorizer.fit_transform(df['text'].apply(lambda x:
' '.join(x)))
title_tfidf = tfidf_vectorizer.fit_transform(df['title'].apply(lambda
x: ' '.join(x)))

#We need inputs (X_train, X_test) and a label (y_train, y_test)
X=txt_tfidf
y=df['label']
X_train, X_test, y_train,y_test =
train_test_split(X,y,test_size=0.3,random_state=101)

print(f"X_train shape: {X_train.shape}")
print(f"y_train shape: {y_train.shape}")

print(f"X_test shape: {X_test.shape}")
print(f"y_test shape: {y_test.shape}")

X_train shape: (14000, 969)
y_train shape: (14000,)
X_test shape: (6000, 969)
y_test shape: (6000,)

#Navie Bayes Model
nb_model = MultinomialNB()
nb_model.fit(X_train, y_train)

#predict
y_pred_nb = nb_model.predict(X_test)

#evaluate the model
print('      Naives Bayes model Summary')
print('\n')
acc_nb = accuracy_score(y_test,y_pred_nb)
print(f"Accuracy: {acc_nb:.2f}")
print('\n')
confusion_matrix(y_test,y_pred_nb)
plt.figure(figsize=(12,8))
sns.set(font_scale=1.2)
sns.heatmap(confusion_matrix(y_test,y_pred_nb),annot=True,fmt='g',cmap
='Blues',
            xticklabels=['Fake', 'True'], yticklabels=['Fake',

```

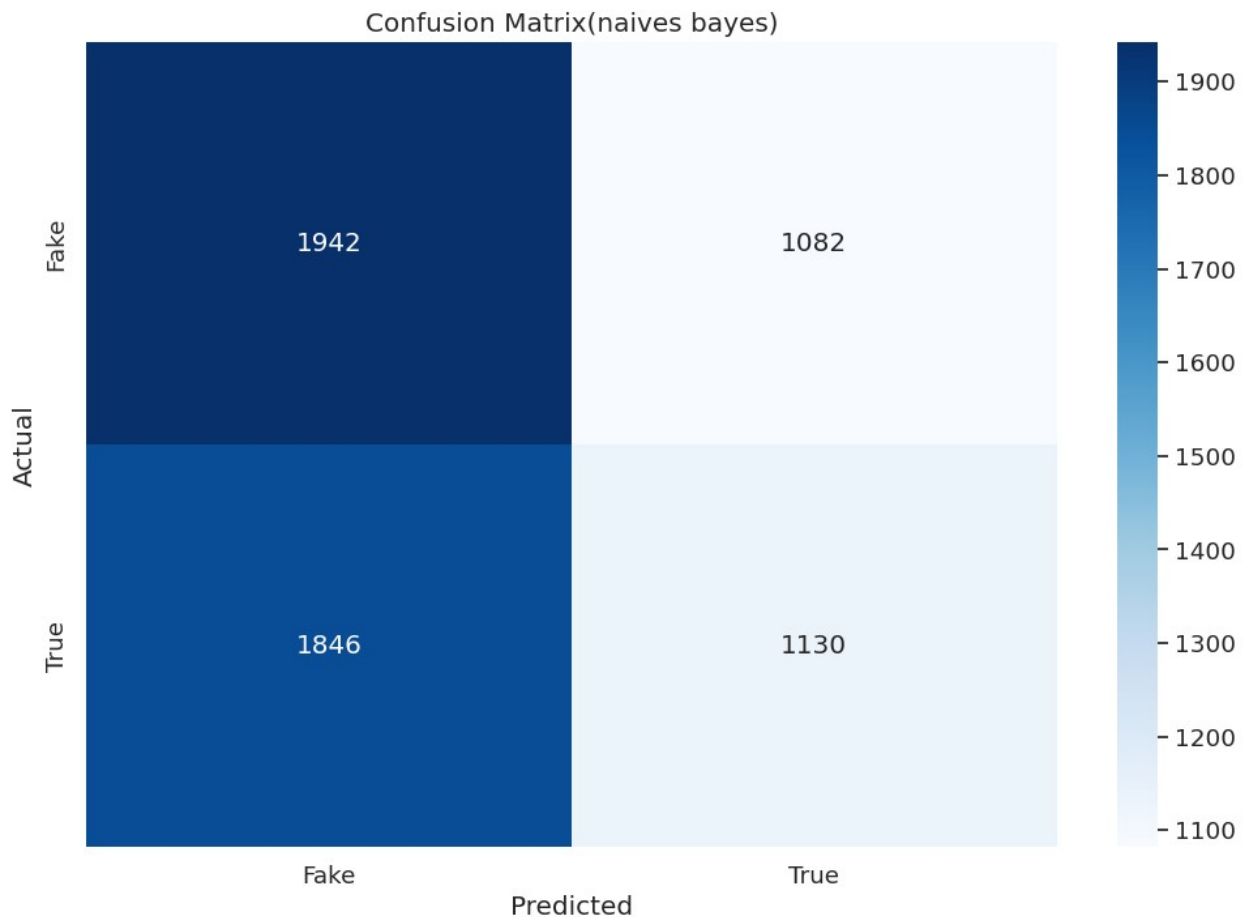
```

'True'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title('Confusion Matrix(naives bayes)')
plt.show()
print('\n')
classification_report_nb = classification_report(y_test, y_pred_nb)
print("Classification Report:\n", classification_report_nb)

```

Naives Bayes model Summary

Accuracy: 0.51



```

Classification Report:
              precision    recall  f1-score   support

fake         0.51         0.64         0.57         3024

```


real	0.51	0.38	0.44	2976
accuracy			0.51	6000
macro avg	0.51	0.51	0.50	6000
weighted avg	0.51	0.51	0.50	6000

#decision tree model

```
dt_model = DecisionTreeClassifier()
dt_model.fit(X_train, y_train)
```

#predict

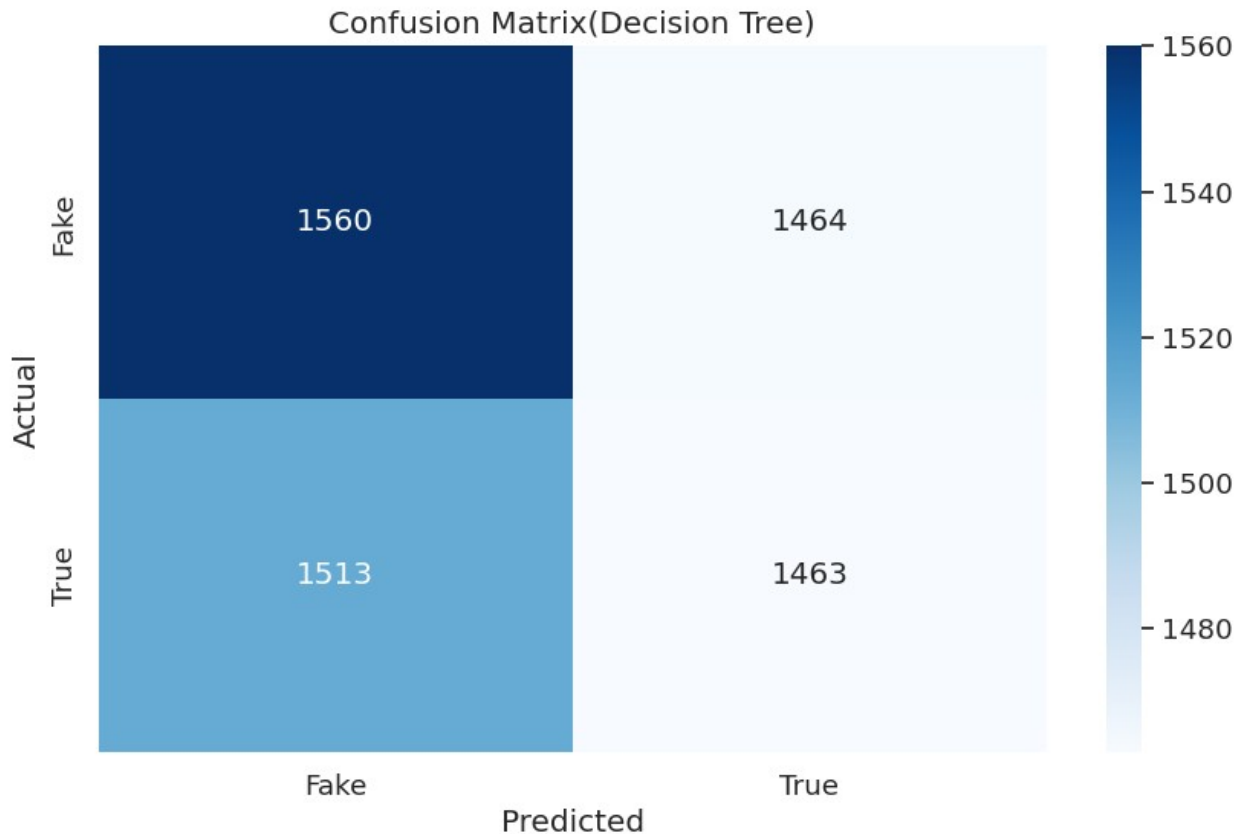
```
y_pred_dt = dt_model.predict(X_test)
```

#evaluate the model

```
print('          Decision Tree model Summary')
print('\n')
acc_dt = accuracy_score(y_test, y_pred_dt)
print(f"Accuracy: {acc_dt:.2f}")
print('\n')
cm_dt = confusion_matrix(y_test, y_pred_dt)
plt.figure(figsize=(10, 6))
sns.set(font_scale=1.2)
sns.heatmap(cm_dt, annot=True, cmap='Blues', fmt='g',
            xticklabels=['Fake', 'True'], yticklabels=['Fake',
            'True'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title('Confusion Matrix(Decision Tree)')
plt.show()
print('\n')
classification_report_dt = classification_report(y_test, y_pred_dt)
print("Classification Report:\n", classification_report_dt)
```

Decision Tree model Summary

Accuracy: 0.51



Classification Report:

	precision	recall	f1-score	support
fake	0.51	0.52	0.51	3024
real	0.50	0.49	0.50	2976
accuracy			0.50	6000
macro avg	0.50	0.50	0.50	6000
weighted avg	0.50	0.50	0.50	6000

```
#random forest classifier
rf_model = RandomForestClassifier()
rf_model.fit(X_train, y_train)
#predict
rf_pred = rf_model.predict(X_test)
#evaluate
print('          Random Forest model Summary')
print('\n')
acc_rf = accuracy_score(y_test, rf_pred)
print(f"Accuracy: {acc_rf:.2f}")
print('\n')
```

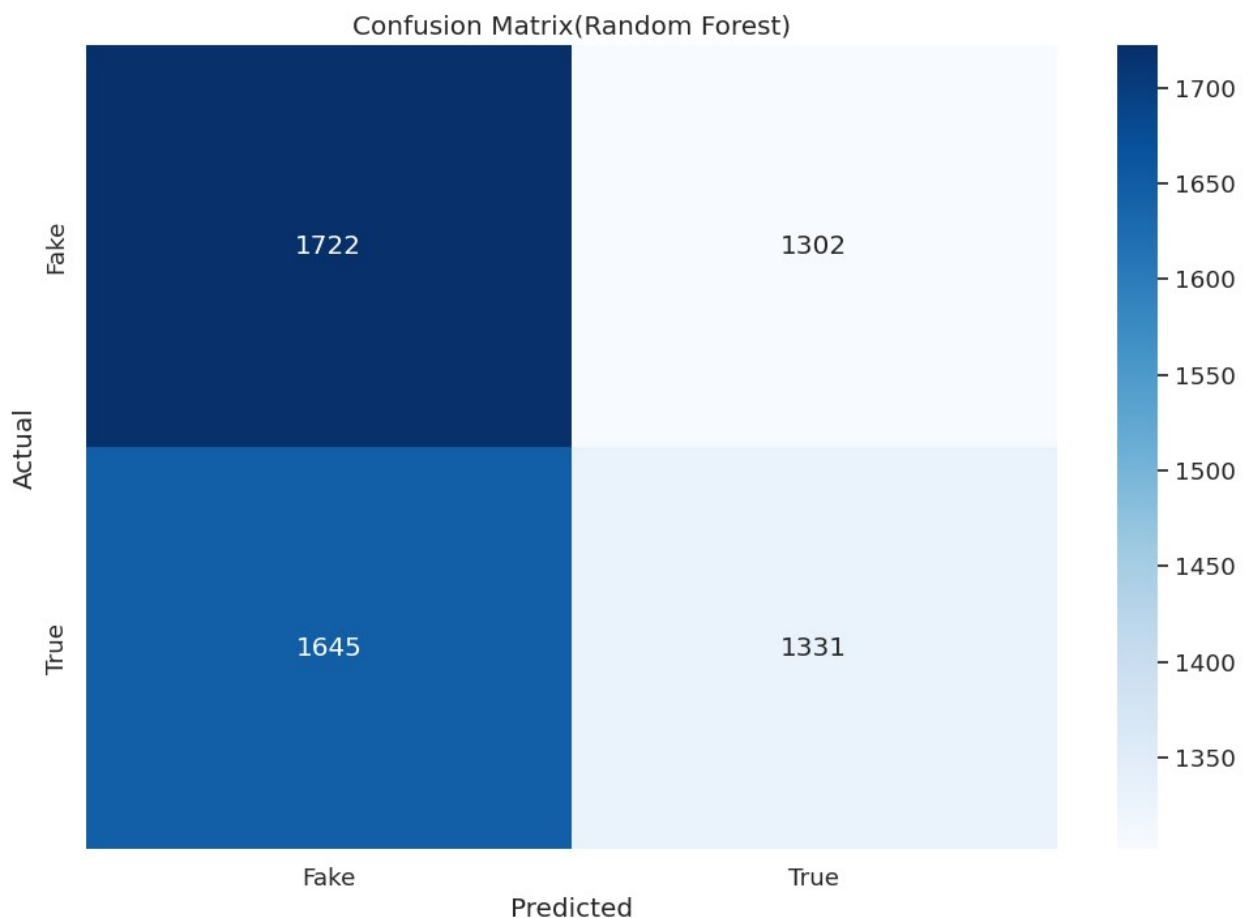
```

cm_rf=confusion_matrix(y_test,rf_pred)
plt.figure(figsize=(12,8))
sns.set(font_scale=1.2)
sns.heatmap(cm_rf,annot=True,cmap='Blues',fmt='g',
            xticklabels=['Fake','True'], yticklabels=['Fake','True'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title('Confusion Matrix(Random Forest)')
plt.show()
print('\n')
classification_report_rf = classification_report(y_test, rf_pred)
print("Classification Report:\n", classification_report_rf)

```

Random Forest model Summary

Accuracy: 0.51



Classification Report:				
	precision	recall	f1-score	support
fake	0.51	0.57	0.54	3024
real	0.51	0.45	0.47	2976
accuracy			0.51	6000
macro avg	0.51	0.51	0.51	6000
weighted avg	0.51	0.51	0.51	6000

```

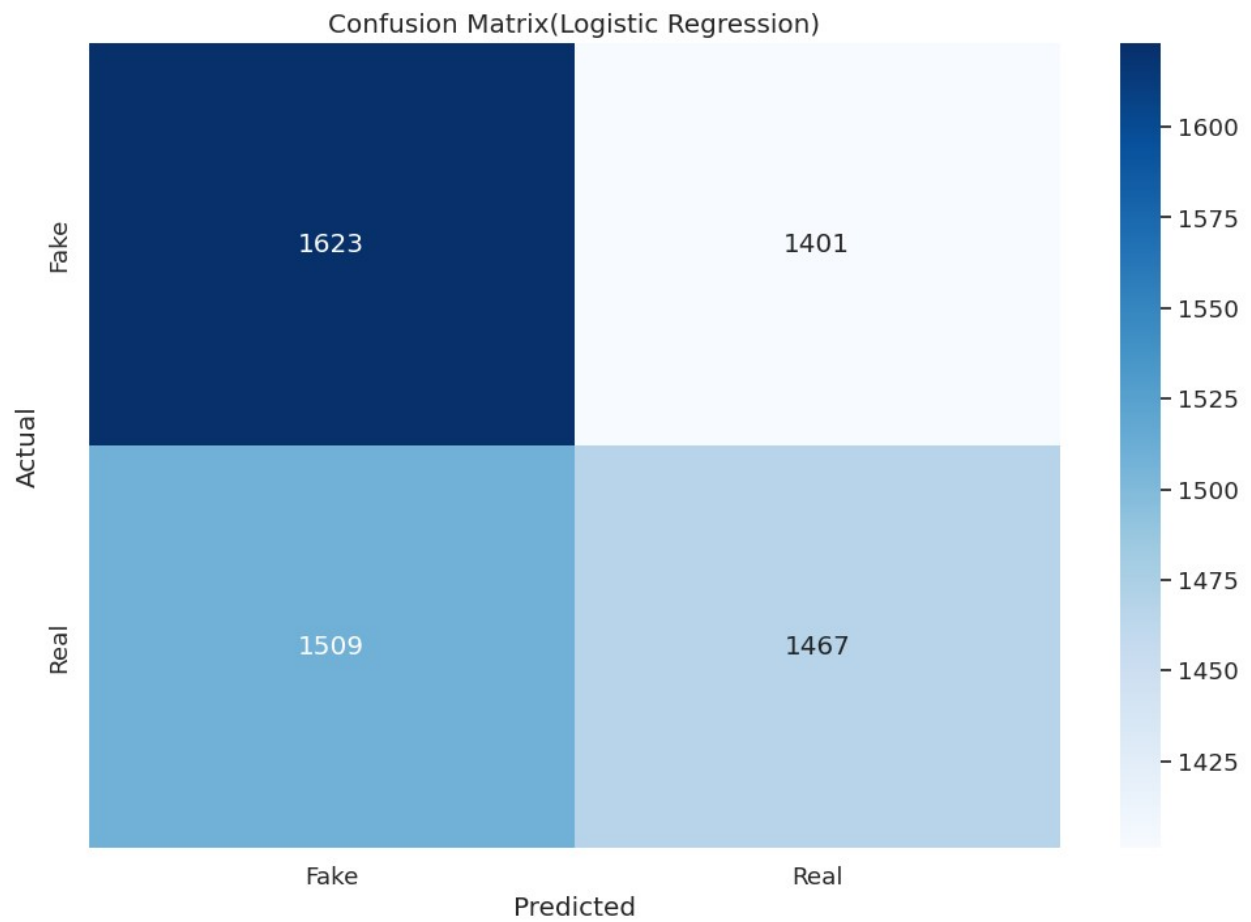
#Logistic Regression model
lr_model = LogisticRegression()
lr_model.fit(X_train, y_train)
#predict
lr_pred = lr_model.predict(X_test)
#evaluate
print('          Logistic Regression model Summary')
print('\n')
acc_lr = accuracy_score(y_test,lr_pred)
print(f"Accuracy: {acc_lr:.2f}")
print('\n')
cm_lr=confusion_matrix(y_test,lr_pred)
sns.set(font_scale=1.2)
plt.figure(figsize=(12,8))
sns.heatmap(cm_lr,annot=True,cmap='Blues',fmt='g',
            xticklabels=['Fake','Real'], yticklabels=['Fake','Real'])
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title('Confusion Matrix(Logistic Regression)')
plt.show()
print('\n')
classification_report_lr = classification_report(y_test, lr_pred)
print("Classification Report:\n", classification_report_lr)
print('\n')
print('\n')
lr_test_precision = precision_score(y_test, lr_pred,pos_label='real')
lr_test_recall = recall_score(y_test, lr_pred,pos_label='real')
lr_test_f1 = f1_score(y_test, lr_pred,pos_label='real')

# Print precision, recall, and F1 score for Logistic Regression
print("Logistic Regression Test Precision:", lr_test_precision)
print("Logistic Regression Test Recall:", lr_test_recall)
print("Logistic Regression Test F1 Score:", lr_test_f1)

```

Logistic Regression model Summary

Accuracy: 0.52



Classification Report:

	precision	recall	f1-score	support
fake	0.52	0.54	0.53	3024
real	0.51	0.49	0.50	2976
accuracy			0.52	6000
macro avg	0.51	0.51	0.51	6000
weighted avg	0.51	0.52	0.51	6000

Logistic Regression Test Precision: 0.5115062761506276

Logistic Regression Test Recall: 0.49294354838709675
Logistic Regression Test F1 Score: 0.502053388090349

```
import nltk
from nltk.corpus import stopwords

def predict_title(title_text):
    # Preprocess the title
    preprocessed_title_text = title_text.lower()
    preprocessed_title_text =
nltk.word_tokenize(preprocessed_title_text)
    stop_words=set(stopwords.words('english'))
    preprocessed_title_text = [word for word in
preprocessed_title_text if word not in stop_words]

    # Convert the preprocessed text into TF-IDF vectors
    tfidf_vector = TfidfVectorizer.transform(["
".join(preprocessed_title_text)])

    # Make the prediction
    prediction = nb_model.predict(tfidf_vector)

    return prediction[0] # Return the string label directly

# Example titles
Prediction_head_1 = "BREAKING: Alien Spaceship Lands in Central Park,
First Contact Imminent"
Prediction_head_2 = "Global Warming Reaches Critical Point, Scientists
Warn of Irreversible Damage"

# Predict and display results for both titles
prediction_1 = predict_title(Prediction_head_1)
prediction_2 = predict_title(Prediction_head_2)

if prediction_1 == 'real':
    print("Title 1: The news is likely true.")
else:
    print("Title 1: The news is likely fake.")

if prediction_2 == 'real':
    print("Title 2: The news is likely true.")
else:
    print("Title 2: The news is likely fake.")

Title 1: The news is likely fake.
Title 2: The news is likely true.

import xgboost as xgb
from sklearn.preprocessing import LabelEncoder

#Encode labels: fake -> 0, real -> 1
```

```

le = LabelEncoder()
y_train_num = le.fit_transform(y_train)
y_test_num = le.transform(y_test)

# Initialize and train XGBoost
xgb_model = xgb.XGBClassifier(n_estimators=200, learning_rate=0.1,
max_depth=6)
xgb_model.fit(X_train, y_train_num)

# Predict and evaluate
y_pred_xgb = xgb_model.predict(X_test)
print(f"XGBoost Accuracy: {accuracy_score(y_test_num,
y_pred_xgb):.2f}")

XGBoost Accuracy: 0.51

```

LSTM model is generally better than TD-IDF even though TD-IDF is fast, efficient. LSTM excels in semantic understanding, whereas TF-IDF is better for keyword-driven tasks.

```

max_words = 10000
maxlen = 200

# Join the tokenized words from the 'text' column back into strings
for the tokenizer
text_data = df['text'].apply(lambda x: ' '.join(x))

tokenizer = Tokenizer(num_words=max_words)
tokenizer.fit_on_texts(text_data)

# Convert text to sequences and pad them
sequences = tokenizer.texts_to_sequences(text_data)
X_pad = pad_sequences(sequences, maxlen=maxlen)

# Re-split the data for the Deep Learning model
# First, get the full numeric labels for the entire dataset (from the
original 'label' column)
y_full_numeric_labels = LabelEncoder().fit_transform(df['label'])

X_train_dl, X_test_dl, y_train_dl, y_test_dl = train_test_split(X_pad,
y_full_numeric_labels, test_size=0.3, random_state=101)

# Building LSTM Model
model = Sequential([
    Embedding(max_words, 128, input_length=maxlen),
    LSTM(64, dropout=0.2, recurrent_dropout=0.2),
    Dense(1, activation='sigmoid')
])

model.compile(optimizer='adam', loss='binary_crossentropy',
metrics=['accuracy'])

```


4. Train

```
model.fit(X_train_dl, y_train_dl, epochs=5, batch_size=32,
validation_data=(X_test_dl, y_test_dl))
```

Epoch 1/5

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/
embedding.py:97: UserWarning: Argument `input_length` is deprecated.
Just remove it.
  warnings.warn(
```

```
438/438 _____ 214s 461ms/step - accuracy: 0.4923 -
loss: 0.6937 - val_accuracy: 0.4915 - val_loss: 0.6939
```

Epoch 2/5

```
438/438 _____ 180s 410ms/step - accuracy: 0.5747 -
loss: 0.6824 - val_accuracy: 0.4847 - val_loss: 0.7073
```

Epoch 3/5

```
438/438 _____ 151s 344ms/step - accuracy: 0.6232 -
loss: 0.6507 - val_accuracy: 0.4827 - val_loss: 0.7312
```

Epoch 4/5

```
438/438 _____ 135s 308ms/step - accuracy: 0.6555 -
loss: 0.6217 - val_accuracy: 0.4827 - val_loss: 0.7547
```

Epoch 5/5

```
438/438 _____ 198s 451ms/step - accuracy: 0.6789 -
loss: 0.5981 - val_accuracy: 0.4923 - val_loss: 0.7927
```

```
<keras.src.callbacks.history.History at 0x78d703963830>
```

#Finding out the most reliable model

#Models applied

```
base = [('nb',MultinomialNB()),
        ('lr',LogisticRegression(max_iter=1000)),

        ('rf',RandomForestClassifier(n_estimators=100,random_state=101))]
```

```
meta_model = LogisticRegression() #final estimator
```

```
stack_model = StackingClassifier(
    estimators=base,
    final_estimator=meta_model,
    cv=5, # Using cross-validation to train the meta-model
    n_jobs=-1
)
```

#train

```
print("Training Stacked Model (this may take a moment)...")
stack_model.fit(X_train,y_train_num)
```

#evaluate

```
y_pred_stack = stack_model.predict(X_test)
stack_acc =accuracy_score(y_test_num, y_pred_stack)
print(f"Stacked Model Accuracy: {stack_acc:.2f}")
```

```

classification_report_stack = classification_report(y_test_num,
y_pred_stack);
print("Classification Report:\n", classification_report_stack)

```

Training Stacked Model (this may take a moment)...

Stacked Model Accuracy: 0.49

Classification Report:

	precision	recall	f1-score	support
0	0.50	0.56	0.53	3024
1	0.49	0.42	0.45	2976
accuracy			0.49	6000
macro avg	0.49	0.49	0.49	6000
weighted avg	0.49	0.49	0.49	6000

#ROC-AUC curve display

```

y_pred_stack_proba = stack_model.predict_proba(X_test)
#second column[:, 1] for the positive class probability
y_scores_real = y_pred_stack_proba[:, 1]
stack_roc_auc = roc_auc_score(y_test_num, y_scores_real)
print(f"Stacked Model ROC-AUC: {stack_roc_auc:.2f}")
print('\n')
print('\n')
plt.figure(figsize=(12, 10))
ax = plt.gca() # Get the current axes
roc_display =
RocCurveDisplay.from_predictions(y_test_num ,y_scores_real ,ax=ax,name
='Stack Model')
plt.plot([0,1],[0,1],color='orange',linestyle='--')
plt.xlabel('False +ve Rate')
plt.ylabel('True +ve Rate')
plt.title('ROC-AUC Curve')
plt.legend()
plt.show()

```

Stacked Model ROC-AUC: 0.49

